INCH-POUND

MIL-S-19500/412A(USAF)
19 June 1990
SUPERSEDING
MIL-S-19500/412(USAF)
8 November 1988

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER
TYPES 2N3846, 2N3847, JANTX, JANTXV, AND JANS

This specification is approved for use by the Rome Air Development Center, Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, power transistors. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (TO-63).

1.3 Maximum ratings.

P_T 1/	P_T 2/	V_{EBO}	V_{CBO}		V_{CEO}		I_C	Safe operating region	T_{STG}	T_{op}
T_A = +25°C	T_C = +100°C		2N3846	2N3847	2N3846	2N3847				
W	W	V dc	V dc	V dc	V dc	V dc	A dc		°C	°C
4	150	10	300	400	200	300	20	See figure 2	-65 to +200	-65 to +200

1/ Derate 26.6 mW/°C linearly to +175°C.

2/ Derate 2 W/°C linearly to +175°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, (AFSC, RADC/RBE-2) Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
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1.4 Primary electrical characteristics.

	h_{FE} V_{CE} = 3 V dc I_C = 10 A dc 1/	$\|h_{fe}\|$ V_{CE} = 10 V dc I_C = 1 A dc f = 1 MHz	V_{BE} V_{CE} = 3 V dc I_C = 10 A dc 1/	$V_{CE(sat)}$ I_B = 1.6 A dc I_C = 10 A dc 1/	C_{obo} V_{CB} = 10 V dc I_E = 0 100 kHz < f < 1 MHz
			V dc	V dc	pF
Min	12	10	---	---	---
Max	60	35	1.2	0.75	750

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

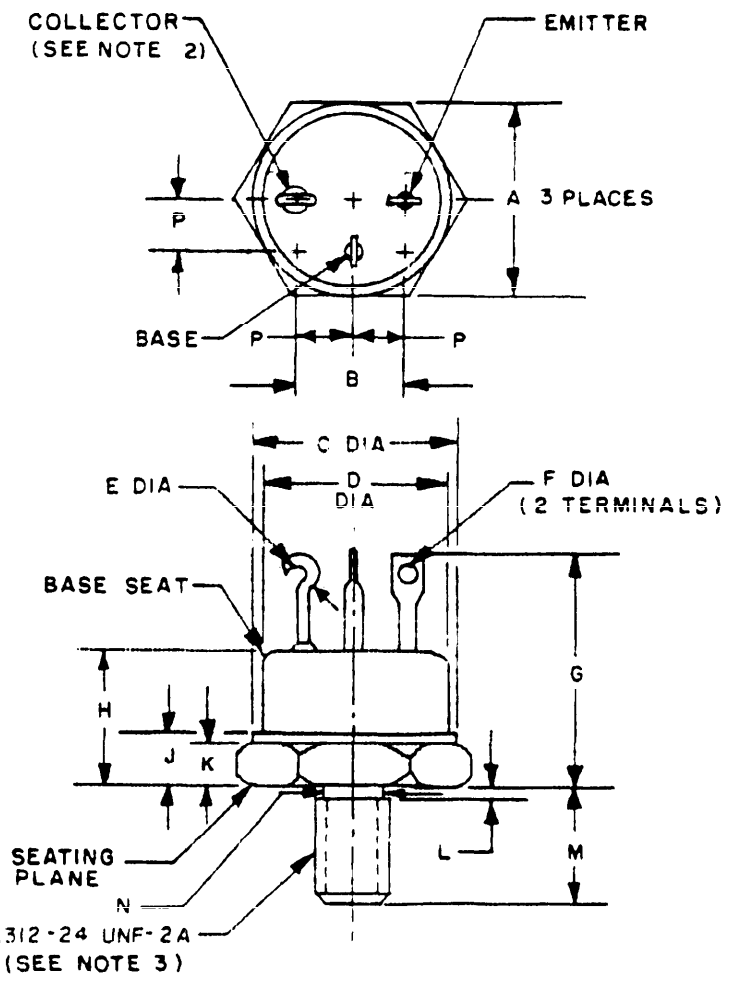
2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exeption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.855	.875	21.72	22.23	
B	.485	.515	12.32	13.08	4, 6
C	.775	.875	19.69	22.23	
D	.745	.775	18.92	19.69	
E	.060	.105	1.52	2.67	2
F	.060	.105	1.52	2.67	
G	.937	1.030	23.80	26.16	7
H	.480	.535	12.19	13.59	
J	---	.300	---	7.62	
K	.090	.167	2.29	4.24	
L	---	.105	---	2.67	8
M	.460	.495	11.68	12.57	
N	.278	.312	7.06	7.92	
P	.240	.260	6.10	6.60	4, 6

NOTES:
1. Metric equivalents are given for general information only.
2. The collector (hook) is electrically connected to the case. The other two terminals shall be electrically isolated from the case.
3. See FED-STD-H28.
4. The orientation of the terminals in relation to the hex flats is not controlled.
5. The case temperature may be measured anywhere on the seating plane within .125 (3.18 mm) of the stud.
6. Terminal spacing measured at base seat only.
7. All three terminals.
8. Maximum unthreaded dimension.

FIGURE 1. Physical dimensions of transistor (TO-63).

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3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be gold-plated, tin-plated, or solder dipped. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500, except at the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, quality assurance provisions, and as otherwise specified herein.

4.3 Screening (JANS, JANTX, AND JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Thermal response, see 4.5.4	Thermal response, see 4.5.4
9	I_{CES} and h_{FE3} of table I shall be measured and the data recorded.	Not applicable
11	I_{CES} and h_{FE3} of table I shall be retested and the data recorded. ΔI_{CES} = 100 percent or 200 µA, whichever is greater. Δh_{FE} = ±20 percent.	I_{CES} and h_{FE3} of table I shall be measured and the data recorded.
12	See 4.3.1	See 4.3.1
13	I_{CES} and h_{FE3} of table I shall be retested and the data recorded. ΔI_{CES} = 100 percent or 200 µA, whichever is greater. Δh_{FE} = ±20 percent.	I_{CES} and h_{FE3} of table I shall be retested and the data recorded. ΔI_{CES} = 100 percent or 200 µA, whichever is greater. Δh_{FE} = ±20 percent.

1/ Shall be performed any time before screen 10.

4.3.1 Power burn-in test conditions. Power burn-in conditions are as follows:

T_J = +187.5 ±12.5°C V_{CB} = 40 ±20 V dc T_A = 30°C ±5°C

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table V herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and tables IIa and IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.4.4 Group E inspection. Group E inspection shall be in accordance with table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Burnout by pulsing. This test shall be conducted in a circuit capable of supplying the specified pulse conditions to the transistor under test. Good engineering practices shall be utilized to eliminate inductive effects in the circuitry and power supply. Rise and fall times of the pulse shall be less than 1/10 of the specified t_p. Noninductive monitoring techniques shall be utilized and the value of any monitoring resistor shall be less than $V_{CC}/20\ I_C$.

4.5.3 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply:

a. I_M measurement current: 10 mA.

b. V_{CE} measurement voltage: 25 V (same as V_H) (minimum).

c. I_H collector heating current: 4 mA (minimum).

d. V_H collector emitter heating voltage: 25 V (minimum).

e. t_H heating time: Steady state (see MIL-STD-750, method 3131 for definition).

f. t_{MD} measurement delay time: 10 to 80 µs.

g. t_{SW} sample window time: 10 µs (maximum).

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071		---	---	---	---
Subgroup 2						
Collector to emitter breakdown voltage	3011	Bias conditon D; I_C = 200 mA dc; I_B = 0 pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N3846				200	---	V dc
2N3847				300	---	V dc
Collector to emitter cutoff current	3041	Bias condition C V_{BE} = 0	I_{CES}			
2N3846		V_{CE} = 300 V dc;		---	2	mA dc
2N3847		V_{CE} = 400 V dc		---	2	mA dc
Collector to emitter cutoff current	3041	Bias condition D I_B = 0	I_{CEO}			
2N3846		V_{CE} = 200 V dc;		---	5	mA dc
2N3847		V_{CE} = 300 V dc		---	5	mA dc
Emitter to base cutoff current	3061	Bias condition D V_{BE} = 10 V dc I_C = 0	I_{EBO}	---	250	uA dc
Forward-current transfer ratio	3076	Pulsed (see 4.5.1) V_{CE} = 3 V dc; I_C = 1 A dc	h_{FE1}	70	---	---
Forward-current transfer ratio	3076	Pulsed (see 4.5.1) V_{CE} = 3 V dc; I_C = 5 A dc	h_{FE2}	40	240	---
Forward-current transfer ratio	3076	Pulsed (see 4.5.1) V_{CE} = 3 V dc; I_C = 10 A dc	h_{FE3}	12	60	---
Base to emitter voltage	3066	Test condition B pulsed (see 4.5.1) V_{CE} = 3 V dc; I_C = 10 A dc	V_{BE}	---	1.20	V dc
Base to emitter voltage (saturated)	3066	Test condition A pulsed (see 4.5.1) I_B = 1.6 A I_C = 10 A dc	$V_{BE(sat)}$	---	1.30	V dc
Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1) I_B = 1.6 A I_C = 10 A dc	$V_{CE(sat)}$	---	0.75	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:						
Collector to emitter cutoff current	3041	Bias condition C; T_A = +150°C; V_{BE} = 0	I_{CES}			
2N3846		V_{CE} = 300 V dc;		---	10	mA dc
2N3847		V_{CE} = 400 V dc		---	10	mA dc
Low temperature operation:						
Forward-current transfer ratio	3076	T_A = -55°C; pulsed (see 4.5.1) V_{CE} = 3 V dc; I_C = 10 A dc	h_{FE}	10	---	---
Subgroup 4						
Magnitude of common-emitter small-signal short circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc f = 1 MHz I_C = 1.0 A dc	$\lvert h_{fe} \rvert$	10	35	---
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 5 A dc f = 1 kHz	h_{fe}	50	250	---
Open circuit output capacitance	3236	V_{CB} = 10 V dc I_E = 0 100 kHz $\leq f \leq$ 1 MHz	C_{obo}	---	750	pF
Turn-on time		I_C = 10 A dc $I_{B(1)}$ = 2 A dc $I_{B(2)}$ = -2 A dc $V_{BE(off)}$ ~ -7.5 V dc R_L = 15 ohms See figure 2	t_{on}	---	4	µs
Turn-off time		I_C = 10 A dc $I_{B(1)}$ = 2 A dc $I_{B(2)}$ = 2 A dc $V_{BE(off)}$ ~ -7.5 V dc R_L = 15 ohms See figure 2	t_{off}	---	7	µs

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Burnout by pulsing (safe operating area - dc)	3005 (3051)	T_C = +100°C; Pre-pulse condition: figure 3 V_{CE} = 0; I_C = 0 Pulse condition: (see 4.5.2)	---	---	---	---
Test 1		V_{CE} = 7.5 V dc I_C = 20 A dc t_p = 1.0 s, 1 cycle				
Test 2		V_{CE} = 200 V dc I_C = 100 mA dc t_p = 1.0 s, 1 cycle				
Test 3		V_{CE} = 58 V dc I_C = 1.0 A dc t_p = 1.0 s, 1 cycle				
Burnout by pulsing (safe operating area - dc) 2N3847 only	3005 (3051)	T_C = +100°C; Pre-pulse condition: figure 3 V_{CE} = 300 V dc I_C = 20 mA dc t_p = 1.0 s, 1 cycle	---	---	---	---
Unclamped inductive sweep	---	T_C = +100°C I_B = 2 A dc I_C = 20 A dc See figure 4				
Clamped inductive sweep	---	T_C = +100°C I_B = 2 A dc I_C = 20 A dc See figure 5				
End points		See table V, steps 1, 3, 7, and 9	---	---	---	---
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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TABLE IIa. Group B inspection for JANS devices.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Physical dimensions	2066	See figure 1
Subgroup 2		
Resistance to solvents	1022	
Solderability	2026	Depth ≥ 0.10 inch; Time = 10 ± 1 second
Subgroup 3		
Thermal shock (temperature cycling)	1051	
Hermetic seal	1071	
Fine leak		
Gross leak		
Electrical measurements		See table V, steps 1, 3, 5, 6, and 8
Decap-internal visual (design verification)	2075	
Bond strength	2037	Test condition A
Die shear	2017	
Subgroup 4		
Intermittent operation life	1037	V_{CB} = 40 ± 20 V dc; P_T = 5 W $T_A \leq +100°C$; $t_{on} = t_{off}$ = 3 minutes minimum for 2000 cycles. No heat sink nor forced air cooling on the devices shall be permitted
Electrical measurements		See table V, steps 1, 3, 5, 6, 8, 9, 11, 12, and 13

See footnote at end of table.

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TABLE IIa. Group B inspection for JANS devices - Continued.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 5		
Accelerated steady-state operation life	1027	V_{CB} = 5 V dc; $P_T \geq 4$ W at $T_A \leq 100°C$ for 96 hours, adjusted as required by the chosen T_A to give an average lot T_J = 275°C ± 12.5°C. Marking legibility requirements shall not apply.
Electrical measurements		See table V, steps 1, 3, 5, 6, 8, 9, 10, 11, and 12
Bond strength (Al-Au die interconnects only)	2037	Test condition A
Subgroup 6		
Thermal resistance	3131	See 4.5.3

1/ For sampling plan, see MIL-S-19500.

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TABLE IIb. Group B inspection for JANTX and JANTXV.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Resistance to solvents	1022	
Solderability	2026	Depth ≥ 0.10 inch; Time = 10 ± 1 second
Subgroup 2		
Thermal shock (temperature cycling)	1051	
Hermetic seal Fine leak Gross leak	1071	
Electrical measurements		See table V, steps 1, 3, 5, and 9
Subgroup 3		
DC intermittent operating life	1037	$t_{on} = t_{off}$ = 3 minutes minimum for 2000 cycles. No heat sink nor forced air coolings on the device shall be permitted
Electrical measurements		See table V, steps 2, 4, 5, 9, 11, and 13
Subgroup 4		
Decap-internal visual (design verification)	2075	
Bond strength	2037	Test condition A
Subgroup 5		
Not applicable		
Subgroup 6		
High temperature (nonoperating)	1032	T_A = +200°C
Electrical measurements		See table V, steps 1, 4, 5, 9, 10, and 11

1/ For sample plan, see MIL-S-19500.

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TABLE III. Group C inspection (all quality levels).

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Physical dimensions	2066	See figure 1
Subgroup 2		
Thermal shock (glass strain)	1056	Test condition B
Terminal strength (tension)	2036	Test condition A Weight = 7 lbs ± 5 oz Application time = 15 s; tubulated leads only
Terminal strength (stud torque)	2036	Test condition D2 Torque = 30 in-lb Application time = 15 s
Terminal strength (terminal torque)	2036	Test condition D1 Torque = 6 in-oz Application time = 15 s; tubulated leads only
Hermetic seal Fine leak Gross leak	1071	
Moisture resistance	1021	
Electrical measurements		See table V, steps 1, 3, 5, and 6 (JANS) and steps 1, 3, 5, 6, and 9 (JANTX and JANTXV)
Subgroup 3		
Shock	2016	
Vibration, variable frequency	2056	
Constant acceleration	2006	
Electrical measurements		See table V, steps 5 and 6 (all quality levels)

See footnote at end of table.

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TABLE III. Group C inspection (all quality levels) - Continued.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 4		
Salt atmosphere (corrosion)	1041	
Subgroup 5		
Not applicable		
Subgroup 6		
DC intermittent operating life	1036	$t_{on} = t_{off} = 3$ minutes minimum for 6000 cycles. No heat sink nor forced air cooling on the device shall be permitted.
Electrical measurements		See table V, steps 3, 6, 9, 10, 11, and 13 (JANS) and steps 2, 4, 5, 9, 11, and 13 (JANTX and JANTXV)

1/ For sampling plan, see MIL-S-19500.

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TABLE IV. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification large lot quality conformance inspection
	Method	Conditions	
Subgroup 1			22 devices, C = 0
Thermal shock glass strain	1056	0°C to +100°C, 100 cycles	
Hermetic seal	1071		
Electrical measurements		See table V, steps 1, 2, and 3	
Subgroup 2			45 devices, C = 0
Steady-state dc blocking life	1039 or 1049	Condition A; 1,000 hrs	
Electrical measurements		See table V, steps 1 and 4	
Subgroup 3		Photos of cross sections shall be submitted in the qualification report. Vendor shall retain duplicate photos.	3 devices, C = 0
DPA	2102		
Subgroup 4			22 devices, C = 0
Thermal resistance	3131	$R_{\theta JC}$ = 0.5°C/W maximum See 4.5.3	
Subgroup 5			22 devices, C = 0
Barometric pressure	1001	Pressure = 8.0 mm Hg time = 60 seconds; normal mounting	
Subgroup 6			10 devices, C = 0
ESD	1020		

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TABLE V. Groups A, B, C, and E electrical end-point inspection measurements.

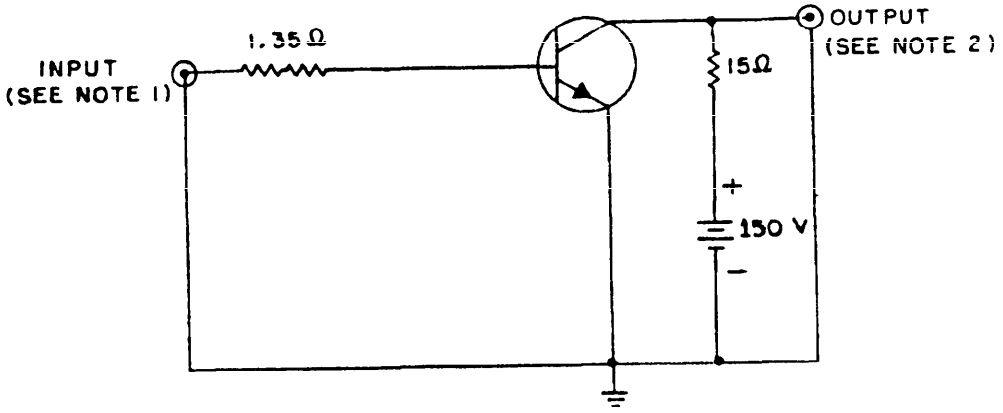
Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Collector to emitter cutoff current 2N3846 2N3847	3041	Bias condition C V_{BE} = 0 V_{CE} = 300 V dc; V_{CE} = 400 V dc	I_{CES}	 --- ---	 2.0 2.0	 mA dc mA dc
2	Collector to emitter cutoff current 2N3846 2N3847	3041	Bias condition C V_{BE} = 0 V_{CE} = 300 V dc; V_{CE} = 400 V dc	I_{CES}	 --- ---	 4.0 4.0	 mA dc mA dc
3	Emitter to base cutoff current	3061	Bias condition D V_{EB} = 10 V dc, I_C = 0	I_{EBO}	---	250	µA dc
4	Emitter to base cutoff current	3061	Bias condition D V_{EB} = 10 V dc, I_C = 0	I_{EBO}	---	500	µA dc
5	Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1) I_B = 1.6 A dc; I_C = 10 A dc	$V_{CE(sat)}$	---	0.75	V dc
6	Base to emitter voltage (saturated)	3066	Test condition A, pulsed (see 4.5.1) I_B = 1.6 A dc; I_C = 10 A dc	$V_{BE(sat)}$	---	1.30	V dc
7	Collector to emitter breakdown voltage 2N3846 2N3847	3011	Bias condition D, pulsed (see 4.5.1); I_B = 0; I_C = 200 mA dc	$V_{(BR)CEO}$	 200 300	 --- ---	 V dc V dc
8	Forward-current transfer ratio	3076	Pulsed (see 4.5.1) V_{CE} = 3 V dc; I_C = 5 A dc	h_{FE2}	40	200	
9	Forward-current transfer ratio	3076	Pulsed (see 4.5.1) V_{CE} = 3 V dc; I_C = 10 A dc	h_{FE3}	12	60	
10	Collector to emitter cutoff current 2N3846 2N3847	3041	Bias condition C V_{BE} = 0 V_{CE} = 300 V dc; V_{CE} = 400 V dc	ΔI_{CES}	100% of initial value or 200 µA whichever is greater		
11	Forward-current transfer ratio	3076	Pulsed (see 4.5.1) V_{CE} = 3 V dc; I_C = 10 A dc	Δh_{FE3}	± 20% change from initial reading		

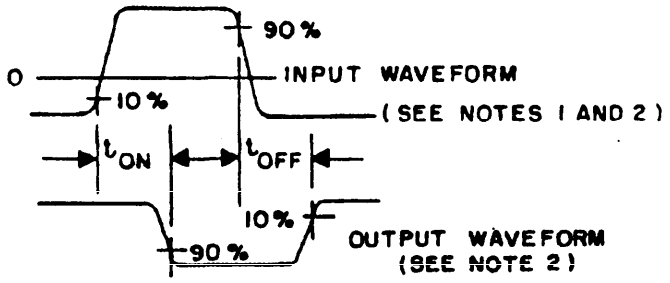
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TABLE V. Groups A, B, C, and E electrical end-point inspection measurements - Continued.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
12	Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1) I_B = 1.6 A dc, I_C = 10 A dc	$\Delta V_{CE(sat)}$	± 50% mV change from initial reading		
13	Thermal response	3131		ΔV_{BE}			

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NOTES:
1. The input waveform has the following characteristics: $t_r \leq 100$ ns, $t_f \leq 100$ ns, PW = 20 µs, duty cycle $\leq 0.2\%$.
2. Waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 5$ ns, $R_{in} \geq 1$ MΩ, $C_{in} \leq 5$ pF.
3. Resistors must be noninductive types.

FIGURE 2. Switching time test circuit.

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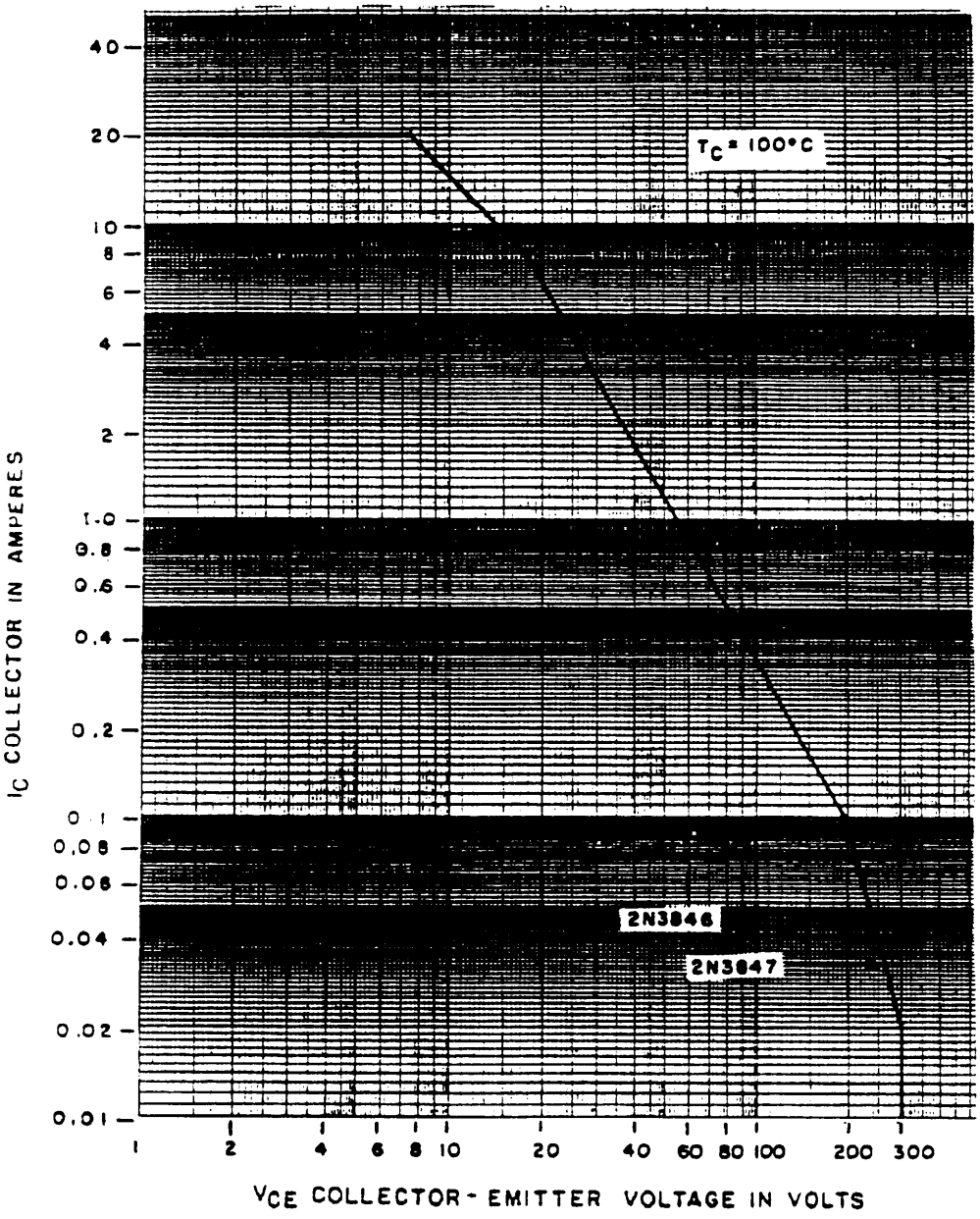
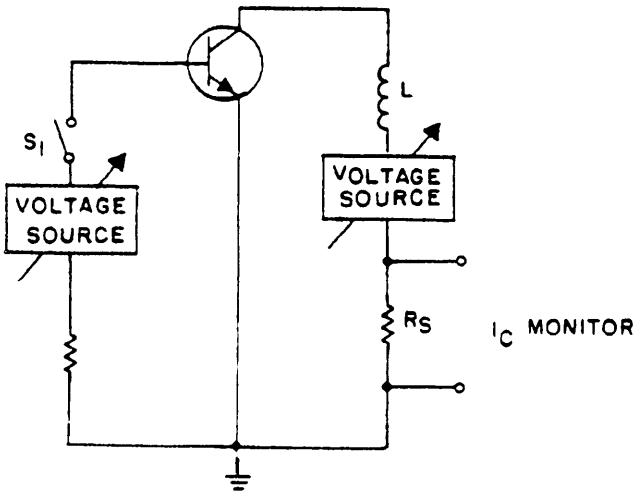


FIGURE 3. Rated safe operating region.

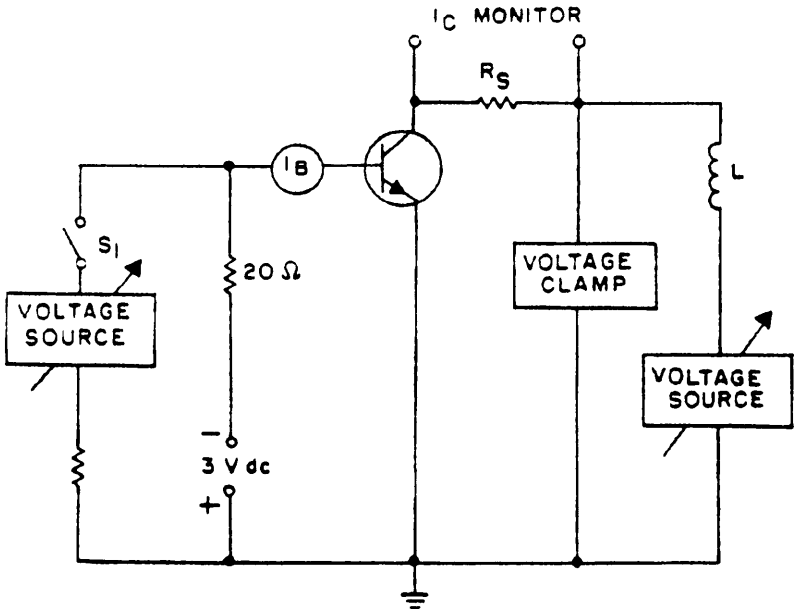
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$R_S \leq 1.0$ ohm (noninductive)
$L = 1.0$ mH

Procedure:
1. With switch S_1 closed, set the specified conditions.
2. Open S_1.
3. Perform specified end-point tests.

FIGURE 4. Unclamped inductive sweep test circuit diagram.



$R_S \leq 1.0$ ohm (noninductive)
$L = 1.0$ mH

Voltage clamp = 2N3846, 300 V dc
2N3847, 400 V dc

Procedure:
1. With switch S_1 closed, set the specified conditions.
2. Open S_1.
3. Perform specified end-point tests.

FIGURE 5. Clamped inductive sweep test circuit diagram.

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4.5.4 Thermal response (ΔV_{BE} measurements). The ΔV_{BE} measurements shall be performed in accordance with MIL-STD-750, method 3131. The ΔV_{BE} conditions and maximum V_{BE} limit shall be derived by each vendor. The ΔV_{BE} measurement and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal response curve shall be plotted. The chosen ΔV_{BE} value shall be considered final after the manufacturer has had the opportunity to test five consecutive lots.

a. I_M measurement current: 10 mA.

b. V_{CE} measurement voltage: 25 V (same as V_H) (minimum).

c. I_H collector heating current: 4 mA (minimum).

d. V_H collector-emitter heating voltage: 25 V (minimum).

e. t_H heating time: 100 ms.

f. t_{MD} measurement delay time: 10 to 80 µs.

g. t_{SW} sample window time: 10 µs (maximum).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead material and finish may be specified (see 3.3.1).

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's part numbers. This information in no way implies that manufacturer's part numbers are suitable as a substitute for the military part number.

Military part number	Manufacturer's CAGE code	Manufacturer's and user's part number
JAN 2N3846	52333 2H784 4U751 6K951 7L707 8S072	2N3846
JAN 2N3847	52333 28784 4U751 6K951	2N3847

6.5 Replacement data. JANTX devices shall be a direct replacement for JAN devices, (example JANTX 2N3846 for JAN 2N3846).

CONCLUDING MATERIAL

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 85
DLA - ES

Preparing activity:
Air Force - 17
Agent:
DLA - ES

(Project 5961-F967)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-19500/412A(USAF)	SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER TYPES 2N3846,2N3847,JAN,JANTX,JANTXV, AND JANS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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